Make Your Move: Platform Assessment Implementation Specification

Document Version: 2.0 (Revised architecture)

Target Audience: CS students (Year 1-4), early-career engineers

Domains Covered: 7 (all domains fully specified)

Nature: Behavior-driven career discovery for software engineering students

1. System Overview

Make Your Move is a behavior-driven career exploration platform. It observes how users think through engineering-style problems and uses those behavioral signals—not self-reported preferences—to surface software domains.

1.1 Core Principles

- **Behavior over self-reporting:** Recommendations are earned through interaction.
- **Transparency over black box:** Users see why they were matched based on session actions.
- **Exploration over labeling:** Results are invitations to investigate, not a verdict.
- **No external tool installs:** All tasks run entirely in-browser.

1.2 Assessment Flow

1. **Context intake:** 3 questions to personalize activity order/weighting.
2. **Behavioural screening (Pre-Assessment):** 6 micro-tasks capturing thinking instincts.
3. **Thinking style reveal:** 3-4 styles surfaced with AI-generated explanations.
4. **In-platform validation tasks:** 2 live tasks per shortlisted style.
5. **Confidence score update:** Rule-based engine recalculates domain weights.
6. **Domain bubble exploration:** Weighted bubbles open to roadmaps and resources.
7. **Personalised learning roadmap:** AI reorders/annotates roadmaps based on skill gaps.

2. Stage 1: Context Intake

Used to personalize activity order, not for domain recommendations.

Question	Options	Effect
Year of study?	Year 1 / 2 / 3 / 4+ / Professional	Year 1-2 see System Flow first; Year 3-4 see Architecture first.
Prior hands-on?	Never / Dabbled / Built something	Prior experience raises the threshold to prevent confirmation bias.
Goal?	No idea / Narrowing / Validating	Exploratory gets wider bubble spread; validators get tighter depth.

3. Stage 2: Pre-Assessment (Behavioural Screening)

Activity 1: System Flow Arrangement

- **Interaction:** Drag 6 unlabelled blocks: `User Request` , `API Gateway` , `Application Server` , `Database` , `ML Model` , `UI Response` .
- **Signals:**
  - `API + DB` prioritized early: **Backend**
  - `ML Model` included logically: **AI/ML**
  - `UI Response` anchored: **Frontend**
  - `Load balancer/monitoring` mentioned in retry: **DevOps**

Activity 2: Messy UI Fix

- **Interaction:** Fix a registration form with scrambled Page title, Email, Password, Error message, Terms, Submit button.
- **Signals:**
  - `Error message` paired to field: **Frontend/Product**
  - `Hierarchy` (Title top, Button bottom): **Frontend**
  - `Grouped related inputs` : **Product Engineering**
  - `Time spent on error message` : **Backend/DevOps** (Reliability instinct)

Activity 3: Live Debug Dashboard

- **Interaction:** Drag pins ('Start here', 'Check next') to panels: `DB log` , `Network tab` , `Server CPU` , `Frontend render` , `Deployment history` .
- **Signals:**
  - `DB query log` first: **Backend**

- `Deployment history` first: **DevOps**
- `Server CPU` first: **DevOps/Backend**
- `Network tab` first: **Backend**
- `Frontend render` first: **Frontend**

Activity 4: Architecture Builder

- **Interaction:** "1 million users" scenario. Select from 9 tiles: `Load balancer`, `App server`, `DB`, `Redis`, `Monitoring`, `Docker`, `ML model`, `Firewall`, `CDN`.
- **Signals:**
  - `Load balancer + monitoring` early: **DevOps**
  - `Redis + DB` prioritized: **Backend**
  - `Firewall` added early: **Cybersecurity** (Strong signal)
  - `CDN` included: **Frontend/DevOps**

Activity 5: Threat Spotter

- **Interaction:** Drag flags to login page vulnerabilities: Plain text password in DOM, No rate-limit, Session token in URL.
- **Signals:**
  - `URL token` flagged before DOM password: **Cybersecurity (High)**
  - `Missing CAPTCHA` flagged: **Backend/Cybersecurity**
  - `Speed to first flag < 8s`: **Cybersecurity (Strong)**

Activity 6: Feature Triage

- **Interaction:** Drag 8 feature cards (Dark mode, 2FA, AI onboarding, etc.) to `Ship`, `Reconsider`, or `Cut`.
- **Signals:**
  - `2FA` shipped unprompted: **Cybersecurity/Product**
  - `In-app notifs` to Needs Thought: **Product Engineering (Strong)**
  - `AI onboarding` to Ship: **AI/ML**
  - `Profile badges` cut: **Product Engineering**

4. Stage 3: Thinking Styles

The engine selects 3-4 styles based on behavior, mapped to 8 specific profiles:

- **The Experience Shaper** (Frontend/Product)
- **The Problem Definer** (Product/AI)
- **The Systems Thinker** (Backend/DevOps)
- **The Reliability Keeper** (DevOps/Backend)
- **The Signal Seeker** (Data/AI)
- **The Hypothesis Tester** (AI/Data)
- **The Threat Anticipator** (Cyber/Backend)
- **The Trust Architect** (Cyber/DevOps)

5. Stage 4: Validation Tasks

In-browser tasks confirming genuine engagement.

- **Frontend/Product:** Tagging motivations (Activity 2) and ranking User Stories.
- **Backend/DevOps:** Live API Triage (JSON analysis + 30-word summary) and Failure Mode Mapping.
- **Data/AI:** Anomaly detection in 20-row table and Confusion Matrix interpretation.
- **Cybersecurity:** Threat Model Builder (User → Browser → API → DB) and Security Code Review (Timing attacks).

6. Stage 5: Scoring Engine (Deterministic)

Pre-Assessment Weights (Samples)

- `Firewall` first in Act 4: **Cybersecurity +4, Backend +1**
- `URL token` flagged in Act 5: **Cybersecurity +4**
- `Deployment history` first in Act 3: **DevOps +4, Backend +1**

Multipliers (Validation Stage)

Validation modifies existing scores:

- **Strong engagement:** x1.8
- **Good engagement:** x1.4
- **Partial/Mixed:** x1.1
- **Low engagement:** x0.9
- **Skipped:** x0.7

7. AI Integration Logic

AI is used **only** for content generation, never for scoring:

1. **Thinking Style Reveal Copy:** Explains why a user matched a style based on their log.
2. **Validation Notes:** One-sentence calibration notes.
3. **Roadmap Personalisation:** Annotates roadmap steps based on observed gaps.

8. Domain Coverage Summary

Domain	Primary Activities	Thinking Style	Validation Task
Backend	1, 3, 4	Systems Thinker	API Triage, Failure Mapping
Frontend	1, 2	Experience Shaper	Component Tag, User Story
DevOps	3, 4	Reliability Keeper	Failure Mode Mapping
Data Eng	1, 4	Signal Seeker	Anomaly Detection
AI/ML	1, 4, 6	Hypothesis Tester	Anomaly, Model Output
Cybersecurity	4, 5, 6	Threat Anticipator	Threat Model, Code Review
Product Eng	2, 6	Problem Definer	User Story Ranking

9. Design Rationale

- **Traceable Scoring:** Users must know "why" a domain appeared; AI language generation is separate from logic.
- **Minimal Typing:** Drag/Rank/Flag actions reduce noise and prevent "faking" through generic text.